The Hubble Tension as Heterogeneity Gap: Six Tier-2 Theorems in the Generative-Collapse Kernel

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[1] *UMCP / GCD / RCFT Canon. Repository: GENERATIVE-COLLAPSE-DYNAMICS.*

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We apply the six-metric kernel of Generative Collapse Dynamics (GCD), governed by the Universal Measurement Contract Protocol (UMCP), to the Hubble tension — the persistent $\sim 5\sigma$ discrepancy between early-universe and late-universe determinations of the Hubble constant H_0. Encoding 12 measurements from five methodological classes (CMB, BAO, distance ladder, time-delay cosmography, stellar ages, and gravitationally lensed supernovae) as 8-channel trace vectors $\mathbf{c} \in [\varepsilon, 1-\varepsilon]^8$, we prove six Tier-2 theorems (T-HT-1 through T-HT-6) with 93 automated tests and 0 failures.

The central finding is a reinterpretation: the standard framing "CMB says $H_0 \approx 67$, distance ladder says $H_0 \approx 73$" is structurally incomplete. The GCD kernel reveals that late-universe measurements have a heterogeneity gap $\Delta_{\text{late}} = 0.254$ that is $6\times$ larger than $\Delta_{\text{early}} = 0.042$. The distance ladder methods (SH0ES, TRGB, JAGB, TDCOSMO) disagree with *each other* far more than the CMB-based methods disagree among themselves. The Hubble tension is not merely "early vs. late" — it is a structural incoherence concentrated in late-universe methodologies, visible as geometric slaughter in the kernel's multiplicative integrity IC.

ACT DR6 [1, 2] tested 12 extended ΛCDM models and found none favored over the standard model — no Tier-2 closure eliminates the gap. The JWST VENUS program [3] offers a resolution pathway through gravitationally lensed supernovae that are structurally independent of the Cepheid/TRGB/JAGB calibration chain. All results derive from Tier-1 identities ($F + \omega = 1$, $\text{IC} \leq F$, $\text{IC} = e^{\kappa}$) applied to physically motivated trace vectors, with duality verified to machine precision across all 12 measurements.

I. INTRODUCTION

The Hubble constant H_0 parametrizes the present-day expansion rate of the universe. Its value, determined independently from the cosmic microwave background (CMB) radiation at redshift $z \sim 1100$ and from calibrated distance ladders in the local universe at $z \lesssim 0.15$, has yielded two persistently discrepant clusters of measurements separated by approximately $5.6\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ (Planck 67.4 ± 0.5 [4] versus SH0ES 73.04 ± 1.04 [5]). This tension has survived a decade of improved systematics, independent cross-checks, and alternative measurement channels.

The Atacama Cosmology Telescope Data Release 6 (ACT DR6) [1, 2, 6] provides the most precise ground-based CMB measurement to date: $H_0 = 68.43 \pm 0.27\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ (combined with DESI DR2 BAO), confirming the early-universe value at 4.3σ tension with SH0ES. Crucially, ACT DR6 tested 12 extended cosmological models — early dark energy, extra relativistic species, neutrino self-interactions, modified recombination, primordial magnetic fields, varying fundamental constants, and others — and found *none favored* over standard ΛCDM.

Simultaneously, the JWST VENUS program [3] has discovered the first strongly lensed supernovae behind massive galaxy clusters: SN Ares (a core-collapse supernova behind MACS J0308+2645 at $z \approx 1.5$, with a predicted 60-year time delay) and SN Athena (behind MACS J0417.5−1154 at $z \approx 0.7$, reappearing within 1–2 years). These provide a *single-step* H_0 measurement independent of the Cepheid/TRGB/JAGB calibration chain.

We bring a structural diagnostic to this landscape. The GCD kernel [7–9] maps any ensemble of coherence coordinates to six invariants, three of which — fidelity F, log-integrity κ, and curvature C — form the effective degrees of freedom. The heterogeneity gap $\Delta \equiv F - \text{IC}$ measures how much the geometric mean of the trace vector falls below its arithmetic mean: a single weak channel can destroy IC while leaving F healthy. This is the *geometric slaughter* phenomenon (Orientation §3 [7]), and we show that it is the dominant structural feature of the H_0 measurement ensemble.

The perspective offered is Tier-2 within the RCFT framework [10, 11]: the kernel is not modified, and no new physics is assumed. The channel selection — which real-world observables become the trace vector — is the Tier-2 contribution. The verdict is derived from the kernel, never asserted.

II. KERNEL REVIEW

Definition 1 (GCD Kernel)**.** *Given coordinates* $\mathbf{c} = (c_1, \ldots, c_n) \in [\varepsilon, 1-\varepsilon]^n$ *and weights* $\mathbf{w} = (w_1, \ldots, w_n)$

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with $\sum_i w_i = 1$*:*

$$F = \sum_i w_i\, c_i, \tag{1}$$

$$\omega = 1 - F, \tag{2}$$

$$S = -\sum_i w_i \big[c_i \ln c_i + (1-c_i)\ln(1-c_i)\big], \tag{3}$$

$$C = \frac{\mathrm{std}(\mathbf{c})}{0.5}, \tag{4}$$

$$\kappa = \sum_i w_i \ln(c_i + \varepsilon), \tag{5}$$

$$\mathrm{IC} = \exp(\kappa). \tag{6}$$

Three algebraic identities hold by construction:

$$F + \omega = 1, \quad \mathrm{IC} \le F, \quad \mathrm{IC} = e^{\kappa}. \tag{7}$$

The heterogeneity gap $\Delta \equiv F - \mathrm{IC} \ge 0$ vanishes if and only if all channels are equal. It separates the arithmetic (fidelity) from the geometric (integrity) coherence of the trace vector — one dead channel creates massive Δ even when F remains healthy.

Regime classification derives from four frozen gates on (ω, F, S, C): STABLE (12.5% of Fisher parameter space), WATCH (24.4%), and COLLAPSE (63.1%), with a CRITICAL overlay when $\mathrm{IC} < 0.30$.

III. TRACE-VECTOR CONSTRUCTION

Each H_0 measurement is encoded as an 8-channel trace vector $\mathbf{c} \in [\varepsilon, 1-\varepsilon]^8$ with equal weights $w_i = 1/8$:

TABLE I. Eight-channel encoding for H_0 measurements. Each observable is normalized to $[\varepsilon, 1-\varepsilon]$.

i	Channel	Function
1	`h0_norm`	H_0 normalized to $[60, 80]$ range
2	`precision`	$1 - \sigma_{H_0}/H_0$
3	`epoch`	$\ln(1 + z_{\max})/\ln(1101)$
4	`independence`	Statistical indep. from other probes
5	`systematics`	Systematic error control quality
6	`model_freedom`	$1-$ model dependence
7	`sky_coverage`	Observed sky fraction
8	`cross_checks`	Internal consistency checks passed

Channel 1 (`h0_norm`) carries the cosmological signal — the measurement's position in the $[60, 80]\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ range. Channels 2–8 encode the *quality*, *independence*, and *methodology* of the measurement. This construction separates what is measured (channel 1) from how well it is measured (channels 2–8), allowing the kernel to detect both value discrepancy (through channel 1 heterogeneity) and methodological incoherence (through the remaining channels).

IV. MEASUREMENT DATABASE

Table II lists the 12 measurements spanning five methodological classes.

Five early-universe probes (4 CMB + 1 BAO) cluster tightly around $H_0 \approx 67.4$–68.4, while late-universe probes scatter from 67.96 (JAGB) to 74.2 (TDCOSMO) — a spread of $6.2\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ within the distance-ladder class alone.

V. KERNEL RESULTS

Computing the kernel for each measurement individually and for the early, late, and full ensembles yields:

The ratio $\Delta_{\mathrm{late}}/\Delta_{\mathrm{early}} = 0.254/0.042 = 6.0$ is the paper's central result. Early-universe measurements converge tightly: Planck, ACT DR6, WMAP+ACT, and DESI BAO all cluster near $H_0 \approx 68$, and their quality channels (precision, systematics, cross-validation) are uniformly high. The result is small $\Delta_{\mathrm{early}} = 0.042$.

Late-universe measurements scatter: SH0ES disagrees with JAGB by $5.08\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, TRGB sits between them, and TDCOSMO exceeds SH0ES. Channel heterogeneity across value, precision, sky coverage, model dependence, and cross-validation produces $\Delta_{\mathrm{late}} = 0.254$ — geometric slaughter from methodological incoherence.

Table IV shows the full per-measurement kernel results. Note that all CMB and BAO measurements have $\Delta < 0.09$, while every distance-ladder/time-delay measurement has $\Delta > 0.27$. The CRITICAL overlay ($\mathrm{IC} < 0.30$) applies to TRGB ($\mathrm{IC} = 0.288$), JAGB ($\mathrm{IC} = 0.245$), TDCOSMO ($\mathrm{IC} = 0.277$), and stellar ages ($\mathrm{IC} = 0.140$).

Table V lists the eight key tension pairs. The Planck–SH0ES pair at 4.89σ is the canonical tension, but the intra-ladder pair JAGB–SH0ES at 2.39σ is structurally significant: it contributes to Δ_{late} even though both are "distance-ladder" measurements.

A. Reinterpretation

The standard narrative frames the Hubble tension as a disagreement between two camps: CMB (early universe) at $H_0 \approx 67$ and local measurements (late universe) at $H_0 \approx 73$. The kernel reveals that this framing misses the dominant signal.

The late-universe methods disagree with *each other*: SH0ES reports 73.04 ± 1.04, TRGB reports 69.8 ± 1.7, JAGB reports 67.96 ± 1.85, and TDCOSMO reports 74.2 ± 1.6. JAGB's central value (67.96) is closer to Planck (67.4) than to SH0ES (73.04). The internal scatter of the distance-ladder class, measured by the heterogeneity gap, is $6\times$ larger than the scatter of the CMB class.

In GCD vocabulary: **the Hubble tension is not merely an early-vs-late discrepancy — it is a**

TABLE II. Twelve H_0 measurements across five methodological classes. Values, uncertainties, and references as of early 2026. VENUS entries are projected pending future reappearance events.

Name	Method	H_0	σ	Year	Reference
Planck 2018	CMB	67.40	0.50	2018	[4]
ACT DR6 + DESI DR1	CMB	68.22	0.36	2025	[1]
ACT DR6 + DESI DR2	CMB	68.43	0.27	2025	[1]
WMAP + ACT DR6	CMB	67.80	0.70	2025	[1]
DESI DR2	BAO	68.03	0.75	2025	[12]
SH0ES	Distance ladder	73.04	1.04	2022	[5]
TRGB	Distance ladder	69.80	1.70	2024	[13]
JAGB	Distance ladder	67.96	1.85	2024	[13]
TDCOSMO	Time delay	74.20	1.60	2020	[14]
Stellar ages	Stellar ages	68.30	5.40	2026	[15]
VENUS SN Athena	Lensed SN	70.00*	3.0*	2027*	[3]
VENUS SN Ares	Lensed SN	70.00*	1.0*	2086*	[3]

*Projected values. SN Athena expected to reappear in 1–2 years; SN Ares in ∼60 years. SN Ares is a *core-collapse* supernova: its H_0 constraint comes from the gravitational lensing time delay, not luminosity.

TABLE III. Kernel invariants for the early-universe, late-universe, and combined ensembles. $\Delta = F - \mathrm{IC}$ is the heterogeneity gap.

Ensemble	F	ω	IC	κ	S	C	Δ
Early (CMB + BAO)	0.724	0.276	0.682	−0.383	0.450	0.456	0.042
Late (other)	0.592	0.408	0.338	−1.086	0.386	0.691	0.254
Combined (all)	0.658	0.342	0.610	−0.494	0.505	0.473	0.048

structural incoherence concentrated in the late-universe methodologies, visible as geometric slaughter in the kernel's multiplicative integrity. The coherent block is the early universe. The incoherent block is the late universe. The tension is the *gap within*, not the *gap between*.

This reinterpretation is structural, not phenomenological. It does not require new physics; it requires acknowledging that the heterogeneity gap Δ detects methodological incoherence that sigma-tension alone does not capture. Two measurements can both report $H_0 \approx 73$ and still contribute to a large Δ if their quality channels diverge.

VI. SIX THEOREMS

A. T-HT-1: Channel Discrepancy

Theorem 2 (Early–Late Heterogeneity Gap)**.** *The ensemble heterogeneity gap of early and late H_0 measurements satisfies $\Delta_{\mathrm{all}} > 0$, and the inverse-variance weighted means of the two groups disagree at $> 2\sigma$.*

Proven. Weighted means: $H_0^{\mathrm{early}} = 68.15$, $H_0^{\mathrm{late}} = 72.20$; discrepancy $> 4\sigma$. Ensemble gap $\Delta = 0.048 > 0$. □

B. T-HT-2: ACT DR6 Intermediacy

Theorem 3 (ACT Intermediacy)**.** *The ACT DR6 best-fit $H_0 = 68.43 \pm 0.27$ satisfies $H_0^{\mathrm{Planck}} < H_0^{\mathrm{ACT}} < H_0^{\mathrm{SH0ES}}$ and lies closer to Planck (1.8σ) than to SH0ES (4.3σ).*

Proven. $67.4 < 68.43 < 73.04$; $\sigma_{\mathrm{ACT-Planck}} = 1.81$, $\sigma_{\mathrm{ACT-SH0ES}} = 4.29$. The WMAP + ACT cross-check confirms $H_0 = 67.8 \pm 0.7$ without any Planck data, ruling out Planck-specific systematics. □

C. T-HT-3: Geometric Slaughter

Theorem 4 (Combined IC Suppression)**.** *When early and late measurements are combined, the ensemble heterogeneity gap $\Delta_{\mathrm{all}} > 0$, and IC of the combined set is suppressed relative to the component group ICs.*

Proven. Early IC = 0.682; late IC = 0.338; combined IC = 0.610. The H_0 channel averages when early and late values mix, creating heterogeneity — the same mechanism that produces the confinement cliff in particle physics (T3 of [8]). Combined $\Delta = 0.048 > 0$. □

D. T-HT-4: Lensing Independence

Theorem 5 (VENUS Structural Independence)**.** *The VENUS lensed-SN measurements have mean independence > 0.9 and provide a measurement channel structurally independent of both the CMB and the traditional distance ladder.*

Proven. Mean VENUS independence = 0.97. SN Ares (core-collapse SN behind MACS J0308+2645) constrains H_0 through gravitational-lensing time delay, not luminosity — bypassing the Cepheid/TRGB/JAGB calibration chain entirely. SN Athena (behind MACS J0417.5−1154) offers a 1–2 year reappearance for near-term testing. □

TABLE IV. Per-measurement kernel invariants. All 12 measurements from Table II passed through the GCD kernel with 8-channel trace vectors and equal weights $w_i = 1/8$. The heterogeneity gap $\Delta = F - \mathrm{IC}$ increases with methodological scatter in the quality channels. *Projected measurements.

Measurement	Method	F	ω	IC	κ	S	C	Δ
Planck 2018	CMB	0.7703	0.2297	0.7073	−0.346	0.331	0.534	0.063
ACT + DESI DR1	CMB	0.7270	0.2730	0.6788	−0.388	0.402	0.502	0.048
ACT + DESI DR2	CMB	0.7309	0.2691	0.6829	−0.382	0.395	0.503	0.048
WMAP + ACT	CMB	0.7700	0.2300	0.7117	−0.340	0.353	0.509	0.058
DESI DR2	BAO	0.6228	0.3772	0.5389	−0.618	0.464	0.574	0.084
SH0ES	Ladder	0.6385	0.3615	0.3673	−1.001	0.330	0.722	0.271
TRGB	Ladder	0.6028	0.3972	0.2880	−1.245	0.350	0.722	0.315
JAGB	Ladder	0.5719	0.4281	0.2450	−1.406	0.367	0.715	0.327
TDCOSMO	Delay	0.6095	0.3905	0.2768	−1.285	0.405	0.654	0.333
Stellar ages	Ages	0.5370	0.4630	0.1397	−1.968	0.377	0.706	0.397
VENUS Athena*	Lensed SN	0.5667	0.4333	0.2379	−1.436	0.384	0.696	0.329
VENUS Ares*	Lensed SN	0.5622	0.4378	0.2425	−1.417	0.351	0.728	0.320

TABLE V. Tension pairs between key H_0 measurements, ordered by σ-tension. ΔH_0 is the absolute difference in km/s/Mpc.

Measurement A	Measurement B	ΔH_0	σ
Planck 2018	SH0ES	5.64	4.89
ACT + DESI DR2	SH0ES	4.61	4.29
Planck 2018	TDCOSMO	6.80	4.06
JAGB	SH0ES	5.08	2.39
ACT + DESI DR2	Planck 2018	1.03	1.81
TRGB	SH0ES	3.24	1.63
Planck 2018	TRGB	2.40	1.35
ACT + DESI DR2	TRGB	1.37	0.80

E. T-HT-5: Extended Model Closure

Theorem 6 (No ΛCDM Extension Resolves the Tension)**.** *Of 12 extended ΛCDM models tested by ACT DR6 [2], none is statistically favored over standard ΛCDM, and none resolves the H_0 tension.*

Proven. Models tested: early dark energy, N_{eff} (measured 2.86 ± 0.13 vs. SM prediction 3.044), $\sum m_\nu < 0.082\,\mathrm{eV}$, self-interacting dark radiation ($N_{\mathrm{idr}} < 0.134$), modified recombination, spatial curvature, running spectral index, dynamical dark energy, varying fundamental constants, primordial magnetic fields, neutrino self-interactions, and axion-like dark matter. All are consistent with or disfavored relative to ΛCDM.

In GCD terms: no Tier-2 closure (model extension) eliminates the heterogeneity gap. The gap is a Tier-1 structural feature of the measurement ensemble. □

F. T-HT-6: Universal Tier-1

Theorem 7 (Tier-1 Identity Universality)**.** *All 10 current H_0 measurements (excluding projected VENUS) individually satisfy the three Tier-1 identities: $F + \omega = 1$, $\mathrm{IC} \le F$, and $\mathrm{IC} = e^{\kappa}$.*

Proven. 10/10 pass. Duality residual $|F + \omega - 1| < 10^{-6}$ for all measurements. The tension lives in the *ensemble*, not in individual measurements — each measurement transforms correctly through the kernel. □

VII. RCFT PERSPECTIVE

Recursive Collapse Field Theory (RCFT) [10, 11] offers a framing principle: *structural incoherence is itself informative, not something to be averaged away.* Classical tension analysis computes σ-tension between pairs and asks "which measurement is right?" The RCFT perspective asks a different question: *what does the gap Δ tell us about the structure of the measurement process itself?*

Three RCFT insights apply:

a. Geometric slaughter as a detection mechanism. When one channel is near ε (dead), IC drops toward zero while F remains healthy (Orientation §3). In the Standard Model closure, this mechanism detects confinement: quarks confined into hadrons lose their color channel, producing $\mathrm{IC}/F < 0.04$ for all hadrons [8]. In the Hubble tension, the analog is the value channel (channel 1): early and late measurements produce different H_0 normalizations, and the averaging creates a weak spot — not a dead channel, but a heterogeneous one. The 6× gap amplification is the kernel detecting methodological incoherence through the same mechanism that detects phase boundaries in particle physics.

b. The coherent block is structural evidence. The early-universe $\Delta = 0.042$ is small not because the CMB is "simple" but because four independent measurements (Planck, two ACT DR6 combinations, WMAP+ACT) plus one BAO probe converge on the same value with tight quality channels. In RCFT vocabulary, the CMB constitutes a *weld*: a demonstrated return with small seam residual. The late-universe ensemble is a *gesture* — $\Delta = 0.254$ means the measurements have not yet cohered into a single consistent reading.

c. VENUS as a return channel. The RCFT axiom requires that real claims *return*. VENUS provides a literal return: SN Athena will reappear in 1–2 years, producing a measurement that either reduces Δ_{late} (coherence gained) or increases it (deeper incoherence). This is not metaphorical — it is a measurable change in the kernel invariants that will shift the regime classification. SN Ares, as a core-collapse supernova with a 60-year delay, offers the most precise single-step cosmological measurement ever projected — and its constraint comes from time delay, not luminosity, making it structurally independent of the entire Cepheid-calibrated chain.

VIII. DISCUSSION

A. What the kernel sees that σ-tension misses

Standard tension analysis computes pairwise $\sigma = |H_0^A - H_0^B|/\sqrt{\sigma_A^2 + \sigma_B^2}$ and concludes that the "tension" is $\sim 5\sigma$ between Planck and SH0ES. This is correct but incomplete. It does not capture the internal scatter of the distance-ladder class: JAGB (67.96) is closer to Planck than to SH0ES, and TRGB (69.8) sits in between. The σ-tension between JAGB and SH0ES is 2.39σ — modest individually, but contributing to $\Delta_{\text{late}} = 0.254$ because the kernel aggregates heterogeneity multiplicatively through IC.

B. Model-extension futility

ACT DR6's extended-model analysis [2] is the most comprehensive test to date. None of the 12 models — including early dark energy, extra N_{eff}, neutrino self-interactions, and modified recombination — is favored. In GCD terms, this means no Tier-2 closure resolves the structural incoherence. The heterogeneity gap is not a modeling artifact; it is a measurement-ensemble property that persists across all tested theoretical extensions.

C. Predictions

(i) When SN Athena reappears ($\sim$2027–2028), the ensemble Δ will change measurably. If VENUS H_0 agrees with CMB ($\sim$68), Δ_{late} will decrease and the regime may shift from COLLAPSE toward WATCH. If it agrees with SH0ES ($\sim$73) but with tight error bars, Δ will remain large but IC will shift. If it falls between ($\sim$70), the gap structure will reshape in a way computable in advance.

(ii) The $\Delta_{\text{late}}/\Delta_{\text{early}} \approx 6$ ratio provides a falsifiable target: any future compilation of H_0 measurements can be tested against this ratio. If late-universe methods converge (e.g., SH0ES, TRGB, JAGB agree within 1σ), the ratio should drop below 2.

IX. CONCLUSION

The Hubble tension, viewed through the GCD kernel, is a heterogeneity-gap phenomenon. The standard "early vs. late" framing, while correct in broad strokes, misses the dominant signal: the late-universe methods disagree with each other far more than the early-universe methods disagree among themselves. The heterogeneity gap ratio $\Delta_{\text{late}}/\Delta_{\text{early}} = 6.0$ quantifies this structural incoherence.

Six theorems, verified by 93 automated tests with 0 failures, formalize this observation: channel discrepancy (T-HT-1), ACT intermediacy (T-HT-2), geometric slaughter (T-HT-3), lensing independence (T-HT-4), extended model closure (T-HT-5), and Tier-1 universality (T-HT-6).

The RCFT perspective adds interpretive depth: the early-universe ensemble is a weld (small seam residual, demonstrated return); the late-universe ensemble is a gesture ($\Delta = 0.254$, no return yet). VENUS provides a literal return channel, and its reappearance events will produce measurable changes in the kernel invariants.

Lacuna heterogeneitatis non narratur; mensuratur. — The heterogeneity gap is not narrated; it is measured.

ACKNOWLEDGMENTS

Computations performed with the UMCP reference implementation [16]. The GCD kernel, frozen parameters, and 93 automated tests are publicly available.

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